

AIRPLANE PERFORMANCE

III Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE02	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
UNIT-I	INTRODUCTION TO AIRPLANE PERFORMANCE							
Role and design mission of an aircraft specification of the performance requirements and mission profile. Importance of performance analysis, estimation and measurements .Scheduled performance and operational performance of aircraft. International standard Atmosphere. Off - design atmosphere. Measurements of air data. Air data computers.								
UNIT-II	AIRPLANE CRUISE PERFORMANCE							
Thrust required and thrust available curves-Minimum drag speed, minimum power speed. Aerodynamic relationships for a parabolic drag polar. maximum & minimum speeds in level flight -Range and Endurance of aircraft with thrust producing engines and with power producing engines. Cruise techniques: constant angle of attack, constant Mach number, constant Mach number methods, comparison of performance. Effect of weight, altitude and temperature on cruise performance.								
UNIT-III	AIRPLANE CLIMB ,DESCENT AND MANEUVER PERFORMANCE							
Importance of climb and descent performance- safety considerations. Climb and descent techniques, generalized performance analysis for thrust producing, power producing and mixed power plants, maximum climb gradient, climb rate. Descent performance in aircraft operations. Effect of wind on climb and descent performance.manoeuvre performance- v-n diagram -level turn-pull up maneuver -pull down manoeuvre								
UNIT-IV	AIRPLANE TAKE-OFF AND LANDING PERFORMANCE AND FUEL PLANNING							
Estimation of take off distances. effect on the take – off distance, of weight wind runway conditions, ground effect. Take off performance safety factors. Estimation of landing distances, the discontinued landing baulked landing air safety procedures and requirements on performance. Fuel Planning : Fuel planning fuel requirements trip fuel, fuel reserves, tankering. Purpose of performance measurements in flight.Flight testing Principle performance variables weight, altitude and ambient temperature (WAT).								
UNIT-V	APPLICATION OF PERFORMANCE DATA							
Performance summary and fleet selection-the block performance, payload – range diagram. Route analysis and optimization. Operational analysis procedure. Operational performance data for flight planning, take off field performance runway correction chart, aircraft datum performance (WAT) chart, determination of the maximum takeoff weight								
Text Books:								
1. Eshelby M. E (2000), <i>Aircraft performance: Theory and Practice</i> , AIAA Education Series, USA. 2. Anderson J. D. (2011), <i>Aircraft Performance and Design</i> , international edition, McGraw Hill, New Delhi.								
Reference Books:								
1. Mc Cormic B. W.(2010), <i>Aerodynamics, Aeronautics and Flight Mechanics</i> , 2 nd edition, Wiley India Ltd. India 2 Yethout (2003), <i>Introduction to Aircraft Flight Mechanics</i> , AIAA Education Series, USA								

COURSE OUTCOMES: STUDENTS SHOULD BE ABLE TO:

1. Compare and contrast between the performance parameters of civil transport and military aircrafts
2. Calculate the range and endurance of a fixed-wing aircraft with either a jet or a propeller-driven propulsion system in straight and level flight and analyze the various types of cruise techniques
3. Illustrate the performance of an aircraft during its climb and descend
4. Evaluate the factors effecting the take-off and landing performance of the aircraft
5. Apply flight mechanics analytical concepts and aircraft performance data to the preliminary design of a new aircraft to meet defined performance requirements